Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP147BT001-V12

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Revision Record

Rev No.	Date	Description
A/0	2024-06-22	First Release

CONTENTS 目录

Revision Record

Contents

ITEM	PAGE
1. General Specifications	4
2. Pin Assignments	5
3. Electrical Specification	6-8
3.1 Absolute Maximum Ratings	6
3.2 Typical Operation Conditions	7
3.3 Backlight Circuit Characteristics	8
3.4 LCD Current Consumption	8
4. Optical Specification	9
4.1 LCM Optical Characteristics	9
4.2 Measurement system	10
5. Mechanical Drawing	11
6. Reliability Test Items	12
7. Packing and Storage Specification (Reference only)	13 -14
8. Announcements	15

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1.General Specifications

No.	Item 项目	Specification 规格	Unit 单位	Remark
1	LCD Size 液晶面板尺寸	1.47	inch	-
2	Panel Type 面板类型	IPS	-	-
3	Resolution 分辨率	172RGBx320	Pixel	-
4	Display Mode 显示模式	Normally Black	-	-
5	Number of Colors 颜色数量	262k	-	-
6	Viewing Direction 视角	ALL	-	Note1
7	NTSC 色彩饱和度	72	-	Typ.
8	Contrast Ratio 对比度	1500	-	Typ.
9	Luminance 亮度	500	cd/m2	Typ.
10	Module Size 模组尺寸	19.39*36.28*1.46	mm	Note1
11	Panel Active Area 可视区域	17.39(H) × 32.35(V)	mm	Note1
12	Pixel Pitch 像素尺寸	33.7(H) × 101.1(V)	um	-
13	Pixel Arrangement 像素排列	RGB Vertical Stripe	-	-
14	Weight 重量	TBD	g	-
15	Driver IC 驱动芯片	JD9853	-	-
16	Driver IC RAM Size 记忆体	RAM	bit	-
17	Light Source 背光源	3white LEDs	-	-
18	Interface 接口方式	4SPI	-	-
19	Operating Temperature 工作温度	-20~+70	℃	-
20	Storage Temperature 存储温度	-30~+80	℃	-

Note 1: Please refer to the mechanical drawing； 注 1：请参照模组图；

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2.Pin Assignments

No.Pin 序号	**Symbol 符号**	**Function 功能描述**
1	LCD_RST	This signal for reset the device
2	TE	Tearing effect signal is used to synchronize MCU to frame memory writing
3	LCD_CS	Chip select input pin("Low" enable).
4	CLK	Serial clock signal in SPI interface
5	SDA	interface data bus
6	GND	Ground(0V)
7	A	LED Positive
8	K	LED Negative
9	VDD	Power supply pin 3.3V
10	TP_INT	This signal for interrupt TP
11	TP_RST	This signal for reset TP
12	TP_SDA	IIC Serial data signal for TP
13	TP_SCL	IIC Serial clock signal for TP
14	GND	Ground(0V)
15	LCD_RS	Data/command selection

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3. Electrical Specification

3.1 Absolute Maximum Ratings

Item 项目	**Symbol**	**Value**	**Unit**	**Remark**
Analog Power Supply Voltage 模拟供电电压	VCI	-0.3~+4.6	V	-
Digital Power Supply Voltage 数字电源电压	VCC	-0.3~+4.6	V	-
I/O Power Supply Voltage I/O端口供电电压	IOVCC	-0.3~3.3	V	-

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3.2 Typical Operation Conditions

Item 项目	Symbol	Min.最小	Typ.典型	Max.最大	Unit
Analog Supply Voltage 模拟供电电压	VCI	2.5	2.8	3.3	V
Digital Supply Voltage 数字电源电压	VCC	2.5	2.8	3.3	V
I/O Supply Voltage 接口电压	IOVCC	1.65	1.8	3.3	V
Input High Voltage 输入高电平	V_{IH}	0.8*IOVCC	-	IOVCC	V
Input Low Voltage 输入低电平	V_{IL}	0	-	0.2*IOVCC	V
Output High Voltage 输出高电平	V_{OH}	0.8*IOVCC	-	IOVCC	V
Output Low Voltage 输出低电平	V_{OL}	0	-	0.2*IOVCC	V

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3.3 Backlight Circuit Characteristics

Item	Symbol	Min.	Typ.	Max.	Unit
LED Current 背光电流	I_B	-	60	-	mA
LED Voltage 背光电压	Vf	2.8	3.0	3.1	V
Power Consumption功耗	P_{BL}	-	-	-	mW

3.4 LCD Current Consumption

<table>
<tr><th>Item</th><th>Symbol</th><th>Typ.</th><th>Max.</th><th>Unit</th></tr>
<tr><td>Full Mode 正常模式</td><td>VCI+IOVCC</td><td>-</td><td>-</td><td>mA</td></tr>
<tr><td colspan="5">测试条件：VCI=2.8V，IOVCC=1.8V；
Interface 驱动类型：行翻转或者列翻转；
TN Type=>All Black Pattern. TN型液晶面板=>黑色画面；
IPS Type=>All White Pattern. IPS型液晶面板=>白色画面；
Temperature：25℃；温度：室温25摄氏度；</td></tr>
<tr><td>Sleep Mode 休眠模式</td><td>VCI+IOVCC</td><td>-</td><td>-</td><td>uA</td></tr>
<tr><td colspan="5">测试条件：VCI=2.8V，IOVCC=1.8V；
DC/DC converter is enabled. Internal oscillator is started and panel scanning is started.
除IC内部晶振和面板扫描外，其他功能都暂停工作；
Temperature：25℃；温度：室温25摄氏度；</td></tr>
</table>

4.Optical Specification

4.1 LCM Optical Characteristics

Item		Symbol	Condition	Min.	Typ.	Max.	Unit
Viewing Angle Range 视角	Left	θ_L	CR≧10	80	85	-	degree
	Right	θ_R		80	85	-	
	Top	θ_T		80	85	-	
	Bottom	θ_B		80	85	-	
Response Time 响应时间		Ton+Toff	$\theta=\Phi=0^\circ$	-	30	40	ms
Contrast Ratio 对比度		CR	$\theta=\Phi=0^\circ$	1000	1200	-	TYP.
Luminance 亮度		L	$\theta=\Phi=0^\circ$	-	50	-	cd/m^2
Color Chromaticity (CIE1931) 色坐标	White	W_x	Normal $\theta=\Phi=0^\circ$	0.26	0.29	0.32	-
		W_y		0.30	0.33	0.36	
	Red	R_x		0.64	0.67	0.70	
		R_y		0.29	0.32	0.35	
	Green	G_x		0.24	0.27	0.30	
		G_y		0.55	0.58	0.61	
	Blue	B_x		0.10	0.13	0.16	
		B_y		0.08	0.11	0.17	
Uniformity 均匀度		U_L	$\theta=\Phi=0^\circ$	TBD	-	-	%
Flicker 闪烁		-	-	≤20%			-

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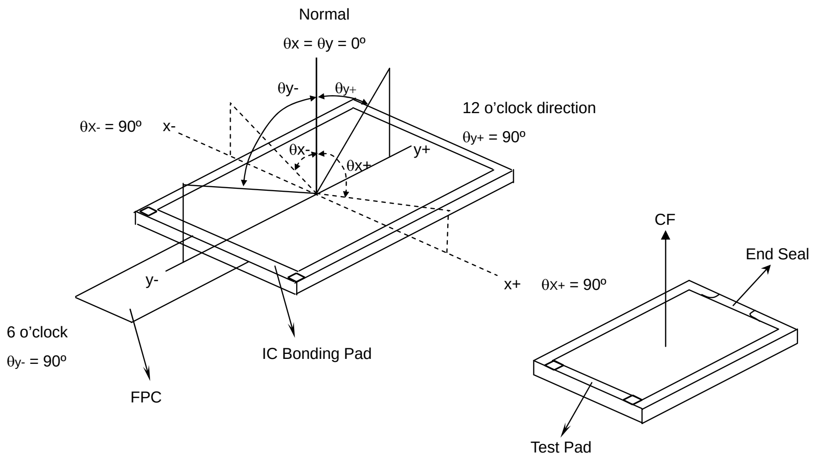
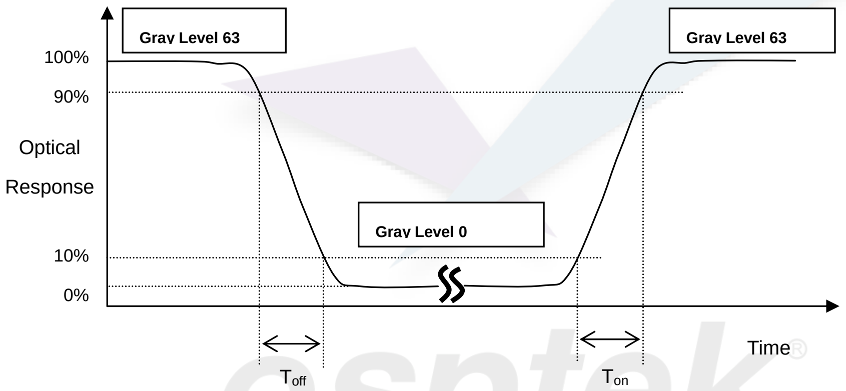
4.2 Measurement system

4.2.1 LCM Viewing Angle


Normal
θx = θy = 0º
θy-
θy+
12 o'clock direction
θy+ = 90º
θx- = 90º
x-
θx-
θx+
y+
CF
End Seal
y-
x+
θx+ = 90º
6 o'clock
θy- = 90º
IC Bonding Pad
FPC
Test Pad

Viewing angle is the angle at which the contrast ratio is greater than 10.The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface.

4.2.2 Response time



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Response time is the time required for the display to transition from white to black (Rising time, Tr) and from black to white (Falling time, Tf) for additional information.

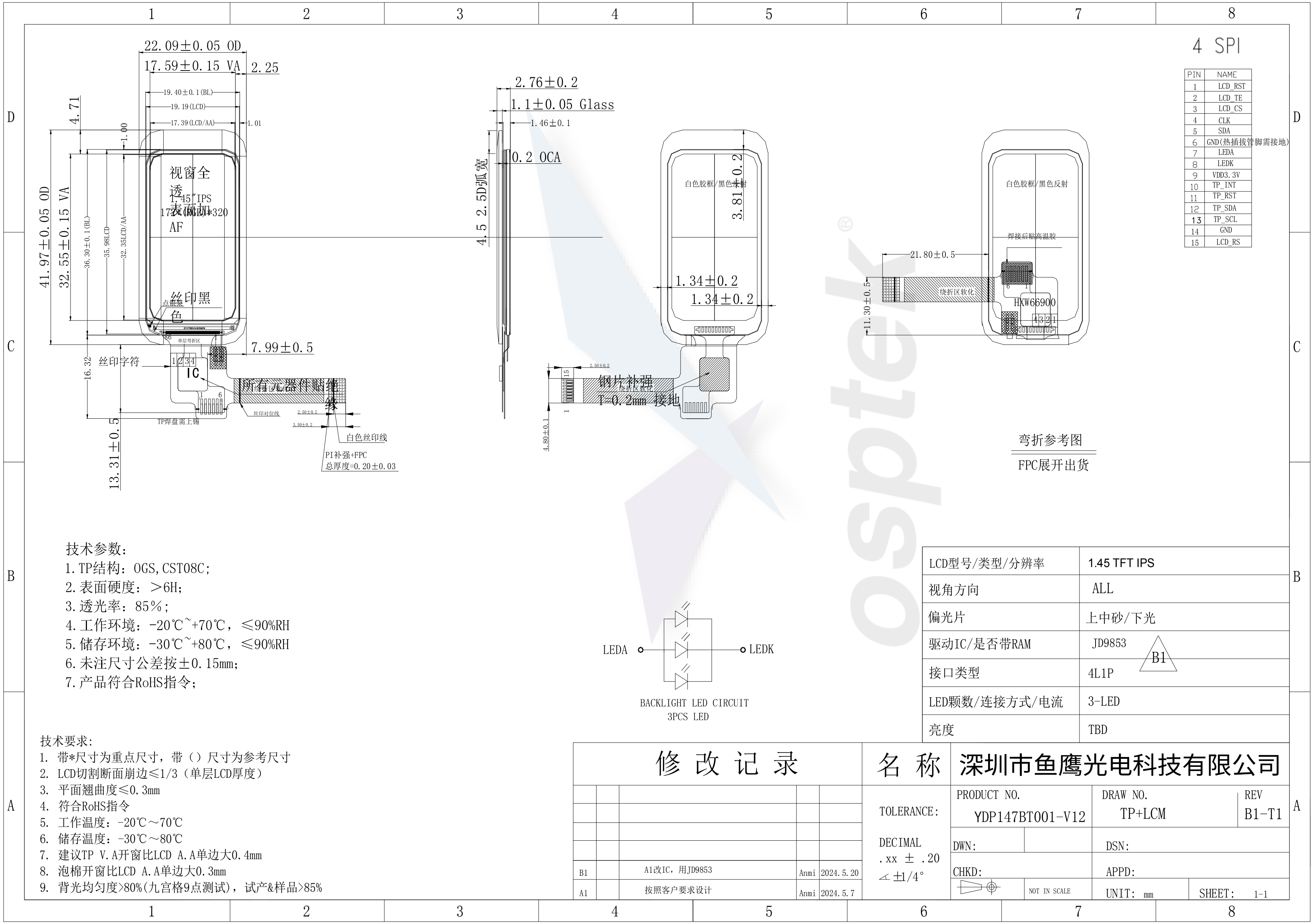
4.2.3 Contrast Ratio (CR)

Contrast Ratio (CR) is defined mathematically as:

$$\text{Contrast Ratio} = \frac{\text{Surface Luminance with all white pixels}}{\text{Surface Luminance with all black pixels}}$$

Surface luminance is the center point across the LCD surface 500mm from the surface with all pixels displaying white.

5. Mechanical Drawing


1
2
3
4
5
6
7
8
A
B
C
D
22.09±0.05 OD
17.59±0.15 VA
2.25
19.40±0.1(BL)
19.19(LCD)
17.39(LCD/AA)
1.01
4.71
1.00
41.97±0.05 OD
32.55±0.15 VA
36.30±0.1(BL)
35.98LCD
32.35LCD/AA
视窗全透
1.45"IPS
172*(RGB)*320
表面加AF
丝印黑色
点银浆
单层弯折区
7.99±0.5
丝印字符
1 2 3 4
IC
16.32
所有元器件贴绝缘
丝印对位线
TP焊盘需上锡
2.50±0.2
3.50±0.2
白色丝印线
PI补强+FPC
总厚度=0.20±0.03
13.31±0.5
2.76±0.2
1.1±0.05 Glass
1.46±0.1
0.2 OCA
4.5 2.5D弧宽
白色胶框/黑色反射
3.81±0.2
1.34±0.2
1.34±0.2
2.50±0.2
15
1
钢片补强
绕折区软化
T=0.2mm 接地
4.80±0.1
白色胶框/黑色反射
焊接后贴高温胶
21.80±0.5
绕折区软化
HXW66900
4 3 2 1
11.30±0.5
弯折参考图
FPC展开出货
LEDA
LEDK
BACKLIGHT LED CIRCUIT
3PCS LED
NOT IN SCALE

4 SPI

PIN	NAME
1	LCD_RST
2	LCD_TE
3	LCD_CS
4	CLK
5	SDA
6	GND(热插拔管脚需接地)
7	LEDA
8	LEDK
9	VDD3.3V
10	TP_INT
11	TP_RST
12	TP_SDA
13	TP_SCL
14	GND
15	LCD_RS

技术参数：
1. TP结构：OGS,CST08C;
2. 表面硬度：>6H;
3. 透光率：85%;
4. 工作环境：-20℃~+70℃，≤90%RH
5. 储存环境：-30℃~+80℃，≤90%RH
6. 未注尺寸公差按±0.15mm;
7. 产品符合RoHS指令;

技术要求：
1. 带*尺寸为重点尺寸，带（）尺寸为参考尺寸
2. LCD切割断面崩边≤1/3（单层LCD厚度）
3. 平面翘曲度≤0.3mm
4. 符合RoHS指令
5. 工作温度：-20℃～70℃
6. 储存温度：-30℃～80℃
7. 建议TP V.A开窗比LCD A.A单边大0.4mm
8. 泡棉开窗比LCD A.A单边大0.3mm
9. 背光均匀度>80%(九宫格9点测试)，试产&样品>85%

LCD型号/类型/分辨率	1.45 TFT IPS
视角方向	ALL
偏光片	上中砂/下光
驱动IC/是否带RAM	JD9853 B1
接口类型	4L1P
LED颗数/连接方式/电流	3-LED
亮度	TBD

修改记录				
B1		A1改IC，用JD9853	Anmi	2024.5.20
A1		按照客户要求设计	Anmi	2024.5.7

名称	深圳市鱼鹰光电科技有限公司		
TOLERANCE:	PRODUCT NO. YDP147BT001-V12	DRAW NO. TP+LCM	REV B1-T1
DECIMAL .xx ± .20 ∠ ±1/4°	DWN:	DSN:	
	CHKD:	APPD:	
	NOT IN SCALE	UNIT: mm	SHEET: 1-1

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6. Reliability Test Items

Test Item 测试项目	**TestCondition 测试条件**	**Test result determinant gist 实验结果判定**
High temperature storage 高温存储	80±3℃，96H；	Inspection after 2~4hours storage at room temperature, the sample shall be free fromdefects: 试验结束后,已测试的LCD样品必须在室内正常温湿度环境下放置2~4个小时以上才能进行功能和外观检查，样品不允许有以下缺陷： 1.Air bubble in the LCD; 模块中有气泡； 2.Non-display; 不显示； 3.Glass crack; 玻璃破碎； 4. The electricalcharacteristicsrequirements shall besatisfied. 需要满足模块电气性能。
Low temperature storage 低温存储	-30±3℃，96H；	
High temperature operation 高温运行测试	60±3℃，96H；	
Low temperature operation 低温运行测试	-20±3℃，96H；	
High temperature /humidity 高温高湿	50℃±3℃,90%±3%RH，96H；	
Thermal Shock 冷热冲击	-30℃/0.5h~+80℃/0.5h for a total 24 cycles；	
Vibration Test 振动测试	Frequency10Hz~55Hz~10HzAmplitude：1.5mm, X，Y，Z directionfor total 1H；(Packing condition)	
ESD test 静电测试	±8KV, Air Mode,150pF/330Ω；	

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Remark: 注意：

1. The test samples should be applied to only one test item.

每个被测试的模块只能用于其中的一个测试项目。

2. Sample size for each test item is 2pcs.

每个测试项目的样品数量为2片。

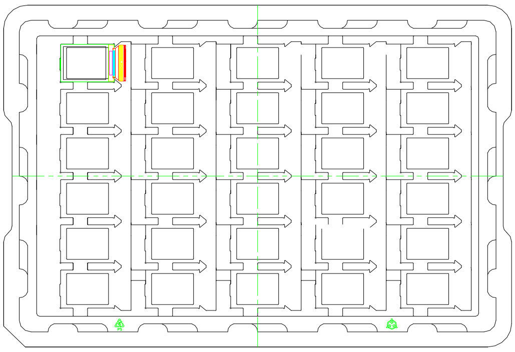
3. Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.

故障判断标准:基本规格,电气特性,机械特性,光电特性。

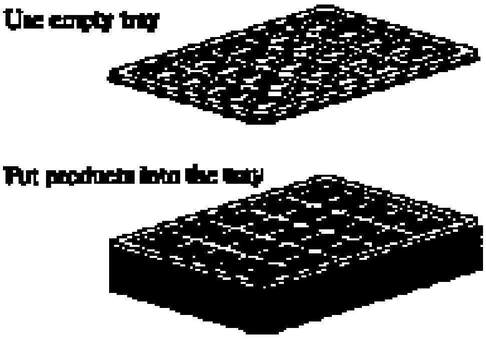
7. Packing and Storage Specification(Reference Only)

7.1 Packing Method

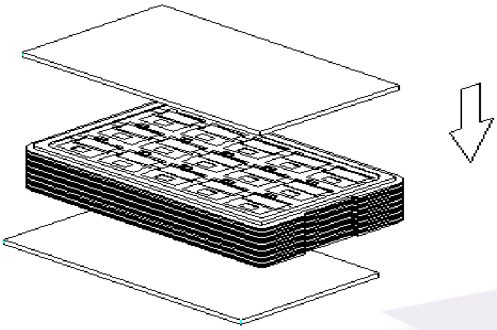
(1)



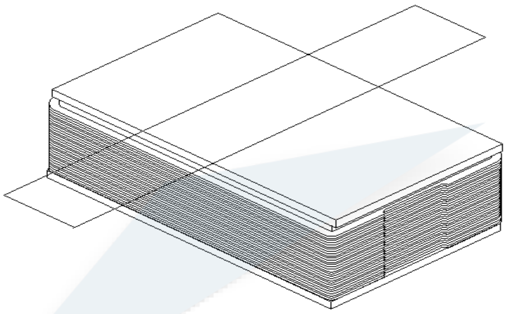
(2)


Use empty tray
Put products into the tray

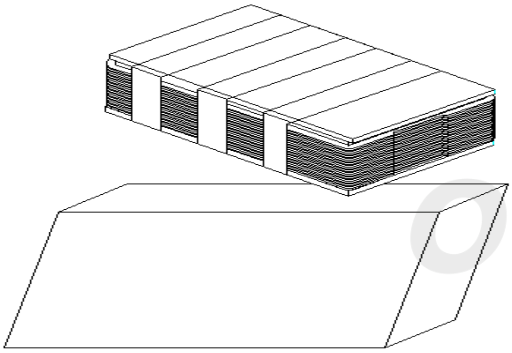
(3)



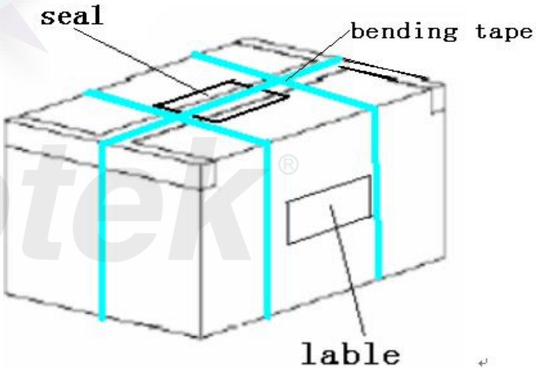
(4)



(5)



(6)


seal
bending tape
lable

1. Put module into tray cavity. .
2. Tray stacking. .
3. Put 1 foam under the tray stack and 1 foam above. .
4. Fix the cardboard to the tray stack with adhesive tape. .
5. Put the tray stack into carton. .
6. Carton sealing with adhesive tape. .

7.2 Storage Method

1. Store in an ambient temperature of 23°C±5°C, and in a relative humidity of 55%±15%. Don't exceed 12 months and expose to sunlight or fluorescent light.
2. Store in a clean environment, free from dust, active gas, and solvent.
3. Store in antistatic container.

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8. Announcements

1. Do not attempt to disassemble or process the LCD module.
2. Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached。
3. Except for soldering the interface, do not make any alterations or modifications with a soldering iron; Ensure welding temperature at 320 ° C to 350 ° C, the welding time control within the 10 s, welding note don't stay too long in the same place to avoid scald FPC.
4. Other matters in not clear before use, please contact our staff to guide.
5. Under the specified electrical parameters, the ambient temperature is 23 ° C ± 5 ° C, the humidity is 55%± 15%, and the hardware is not damaged. The normal service life is 30,000-50,000 hours.

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